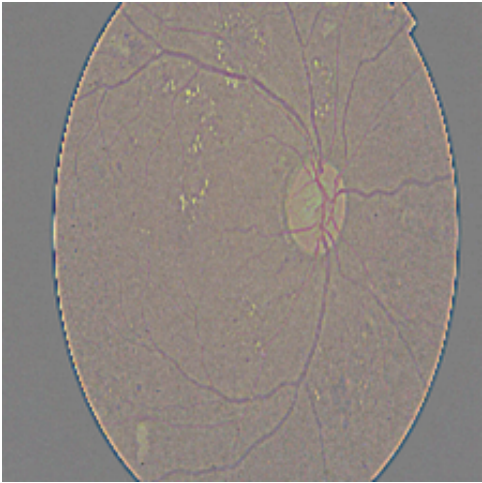


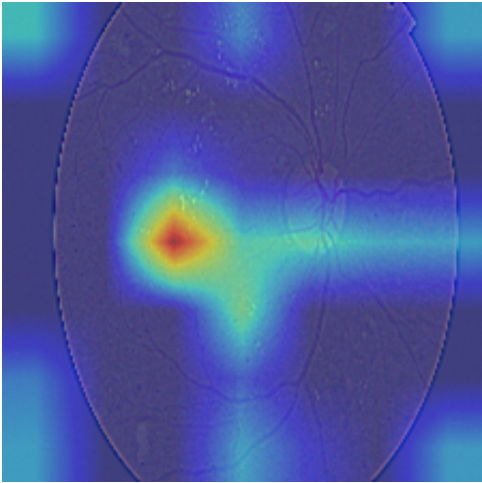
Clinical Diagnostic & RAG Report - Nafld

Target Domain:	Nafld	Image File:	PIL_Image
Predicted Class:	normal	Confidence:	54.1%
95% Conformal Set:	['fatty_liver', 'normal']	Uncertainty Status:	[WARNING] Human Review Recommended

Visual Diagnostic Imaging & Grad-CAM Heatmap



Original Medical Scan



Grad-CAM Saliency Overlay (ROI)

AI Clinical Explanation (FLAN-T5 Grounded RAG)

'Magnetic resonance imaging (MRI) is a noninvasive diagnostic method for the diagnosis of nonalcoholic fatty liver disease (NAFLD) and can be used to differentiate the pathological stages of NAFLD and its complications.'

Faithfulness & Quantitative Text Metrics

NLI Faithfulness Score:	0.5677 / 1.0000	ROUGE-1:	0.8621
BLEU-4:	0.2414	BERTScore Similarity:	0.4934

Retrieved PubMed Literature Evidence

[1] [New advances in diagnosis of nonalcoholic fatty liver disease]. (PMID: 26983472)

'Nonalcoholic fatty liver disease (NAFLD) is the most common chronic liver disease, and the spectrum of this disease includes simple liver steatosis, nonalcoholic steatohepatitis, liver fibrosis, and liver cirrhosis. At present, liver biopsy is still the gold standard for the diagnosis of NAFLD and can determine the degree of steatosis, inflammation grade, and fibrosis stage, but its clinical application is limited by its invasiveness and static nature. New diagnostic markers and noninvasive diagnostic methods have been reportedly recently, and the constant development of magnetic resonance imaging, transient elastography, and controlled attenuation parameter provides new methods for the noninvasive diagnosis of NAFLD.'

[2] NAFLD Is Associated With Quiescent Rather Than Active Crohn's Disease. (PMID: 37454277)

'BACKGROUND AND AIMS: Crohn's disease (CD) confers an increased risk of nonalcoholic fatty liver disease (NAFLD), but the pathogenesis remains poorly understood. We determined if active intestinal inflammation increases the risk of NAFLD in patients with CD. METHODS: Two cohorts (2017/2018 and 2020) with CD and no known liver disease were enrolled consecutively during staging magnetic resonance enterography.'

[3] Application of artificial intelligence techniques for non-alcoholic fatty liver disease diagnosis: A systematic review (2005-2023). (PMID: 38008040)

'BACKGROUND AND OBJECTIVES: Non-alcoholic fatty liver disease (NAFLD) is a common liver disease with a rapidly growing incidence worldwide. For prognostication and therapeutic decisions, it is important to distinguish the pathological stages of NAFLD: steatosis, steatohepatitis, and liver fibrosis, which are definitively diagnosed on invasive biopsy. Non-invasive ultrasound (US) imaging, including US elastography technique, and clinical parameters can be used to diagnose and grade NAFLD and its complications.'

[4] NAFLD Is Associated With Quiescent Rather Than Active Crohn's Disease. (PMID: 37454277)

'Definitive or borderline steatohepatitis was present in most patients that underwent liver biopsy. CONCLUSIONS: Quiescent CD is associated with risk of NAFLD. These findings suggest potentially distinct pathogenic mechanisms of NAFLD in patients with CD compared with the prevailing leaky gut hypothesis proposed for individuals without inflammatory bowel disease.'

[5] Quantitative Diagnosis of Nonalcoholic Fatty Liver Disease with Ultrasound Attenuation Imaging in a Biopsy-Proven Cohort. (PMID: 37407373)

'This study included 60 patients who had clinical suspicion of NAFLD and were referred for liver biopsy after ATI and controlled attenuation parameter (CAP) examinations between September 2020 and December 2021. The histologic hepatic steatosis was graded. The area under curve (AUC) analysis was performed.'